

Renewable energy

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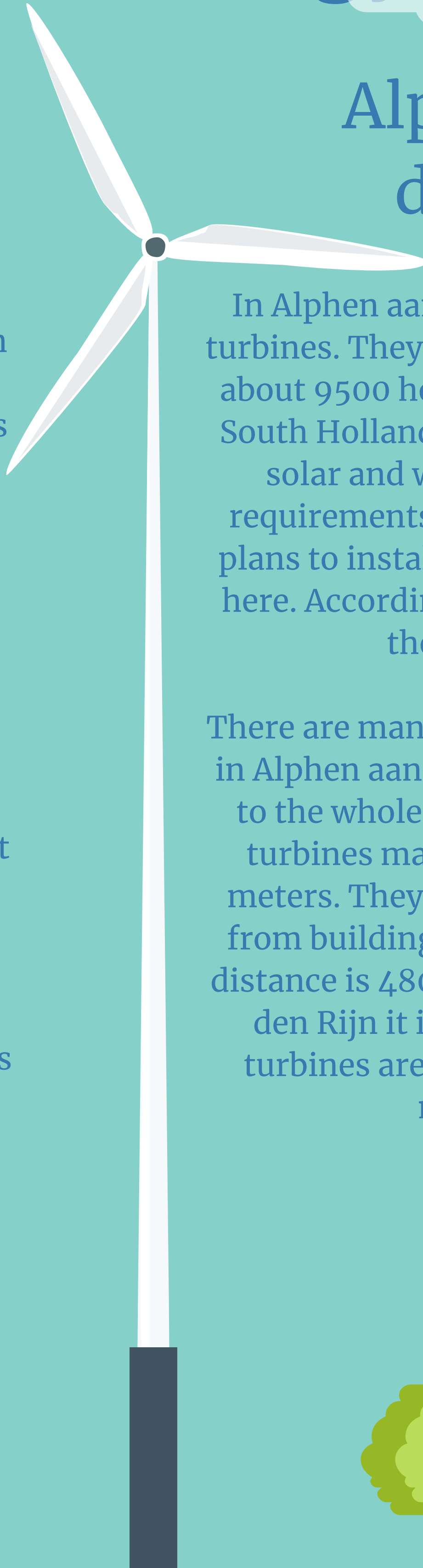
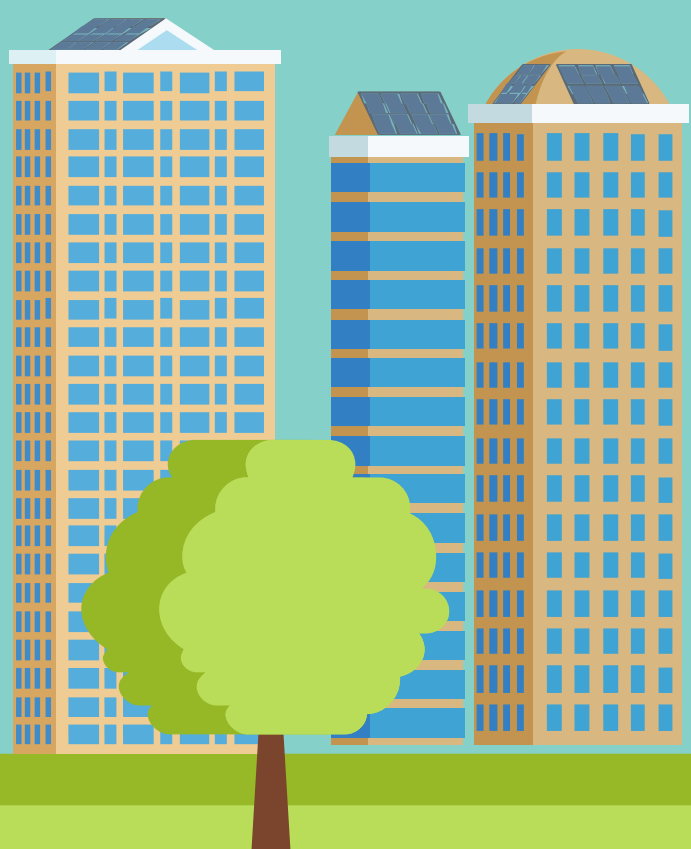
In Istanbul, we have nearly 100 wind turbines that produce enough energy for about 400,000 households. However, the city does not produce enough renewable energy to meet its massive demand, which is why authorities plan to install more turbines in suitable northern areas.

There are strict rules for these projects; while national standards apply, turbine heights in Istanbul are specifically limited by airport flight safety regulations. Additionally, the required distance from buildings is extended to 1,000 meters due to the city's high population density, and no turbines are allowed in protected natural areas like the Northern Forests

Alphen aan den Rijn

In Alphen aan den Rijn we have 4 wind turbines. They produce enough energy for about 9500 households. The province of South Holland does not produce enough solar and wind energy to meet the requirements. That's why the province plans to install even more wind turbines here. According to them, this area is the most suitable.

There are many rules about wind turbines in Alphen aan den Rijn. Some rules apply to the whole of the Netherlands. Wind turbines may not be higher than 240 meters. They must also be at a distance from buildings. In the Netherlands this distance is 480 meters, but in Alphen aan den Rijn it is 550-700 meters. Wind turbines are not allowed in protected natural areas.



Renewable energy

Wind energy

Conclusion

Alphen aan den Rijn and Istanbul possess different yet complementary strengths in renewable energy. Alphen has mastered the "democratization of energy" by involving the local community in energy cooperatives and successfully bringing production down to the neighborhood level. Conversely, Istanbul stands out with its high industrial speed, rapidly modernizing massive industrial basins and public infrastructure to integrate large amounts of clean energy into the grid.

In terms of development, Alphen can further balance its energy portfolio by better integrating wind power with its solar achievements. Istanbul needs to increase its individual residential solar usage, which currently remains at 5% due to urban density, and develop new models to involve households in energy production. Supporting the city's industrial success with residential energy independence is a strategic priority.

These two cities can draw valuable lessons from each other; Istanbul can adopt Alphen's model of social participation to encourage citizens to gain direct economic benefits from energy. Meanwhile, Alphen can adapt Istanbul's high-capacity transformation speed seen in logistics centers and transport networks to its own industrial areas. By utilizing hybrid solutions that combine wind and solar, both cities can meet in a shared vision for a sustainable future.

Wind turbines Istanbul



Wind turbines Alphen aan den Rijn



Renewable energy

Solar energy



Istanbul

As the industrial heart of Turkey, Istanbul is a strategic hub for solar energy. Industrial zones like İkitelli, Tuzla, and Dudullu are transforming factories into clean energy hubs. While urban density limits residential use to around 5%, the vast rooftops of industrial facilities are driving the megacity's renewable energy capacity.

This transformation creates significant value through concrete data. Approximately 5,000 solar panels on a single logistics center or factory roof generate enough power for 800 households annually. This allows factories to lower costs while feeding surplus electricity back into the city grid.

Expanded by public and industrial partnerships, this capacity—including metro stations and bus depots—now powers tens of thousands of homes. Through these investments, Istanbul's industrial strength is gaining a green identity by preventing thousands of tons of carbon emissions each year.



Alphen aan den Rijn



Solar energy is the most common form of renewable energy in Alphen aan den Rijn. More solar energy is generated than wind energy, for example.

The Netherlands has worldwide the most solar panels per person. Each Dutch person has, on average, 3.5 solar panels installed. A total of 3 million households have solar panels. This means that 35% of the residents in the Netherlands have solar panels. In Alphen aan den Rijn 18% of the residents have them. This means that Alphen aan den Rijn is lagging behind the Dutch average.

There is an organization called Zon op Alphen (Sun on Alphen). They have set up several projects to generate even more solar energy. They have placed 1,844 solar panels on the roof of a large company called Rijnvicus. With these panels they can provide enough energy for about 200 households.



Renewable energy



Solar energy

Conclusion

Statiscically, if we look at the ways of the solar energy income Istanbul and Alphen aan den Rijn have a major difference. In Istanbul Solar energy mainly comes from logistic centers and factory roofs while in Alphen aan den Rijn solar energy mainly comes from a households roof. But in Alphen aan den Rijn they also have a few projects which are mainly focused on generating energy for future sale.

So in Alphen aan den Rijn people are mainly dependent on their own renewable energy, while in Istanbul, they are mainly dependent on energy which is generated by the government. This difference mainly occurs because the two countries have very a different public policy. For example, in Alphen aan den Rijn the government is more focused on individual investings for solar energy. While in Istanbul, the government is mainly focused on social investments

The two cities both have very prominent plans for the future, but we noticed a few changes which would be worth it. In Alphen aan den Rijn, the government is ofcourse mainly focused on individual solar energy investments but it would be better they also focused a bit more on social investments, this way they can be more prepared for unexpected future events which may involve more energy than planned. In Istanbul, the government could be more focused on individual solar energy investments. The government is ofcourse producing a lot of energy for the entire city but they could produce even more if the solar panels were spread across the city and not just loads on one factory building for example.

Solar panels Istanbul



Solar panels Alphen Aan Den Rjin



Renewable energy



Future plans



Istanbul

Istanbul is accelerating its renewable energy investments in line with its goal of becoming a "Carbon Neutral" metropolis by 2050. As the most vital part of this strategy, the "Solar City Istanbul" vision aims to transform the rooftops of all public buildings, parking lots, and logistics centers into full-capacity solar power plants.

To utilize the city's limited space efficiently, there are plans to expand "Floating SPP" systems installed on drinking water reservoirs. Additionally, the installation of massive battery storage systems in industrial zones aims to provide the solar energy generated during the day to industrial lines and the city grid uninterruptedly at night.

Within the scope of future plans, it is a priority to power Istanbul's electric public transport fleet entirely from renewable sources and to establish neighborhood-based smart microgrids. Through this comprehensive transformation, Istanbul aims to become not only Turkey's industrial heart but also one of the world's leading green technology metropolises.



Alphen aan den Rijn

In Alphen aan den Rijn we are very focused on our future since it is not only important to us, but also for our future generations. We want to be as careful as possible when it comes to energy usage which is why we have a program which is called "Programma energietransitie" that keeps track of our energy usage and explains multiple energy efficient ways.

This program has four main topics, isolation and save energy, generate and energy system, sustainable heating and lastly sustainability of business parks these four topics are mainly focused on sustainable and frugal/careful ways to use energy

With this program we have a few achievable goals for the future, in 2050 we plan to not use any more natural gas because it is damaging for our climate. We also want 80% of the region to be fully dependent on renewable energy.



Renewable energy



Future plans



Conclusion

According to their future plans, Istanbul seems to have a more aggressive and technology-driven roadmap. While Alphen aan den Rijn focuses on 80% renewable dependency, Istanbul aims for a total "Carbon Neutral" status by 2050 with massive infrastructure projects. Istanbul's plans to convert all public rooftops into power plants and electrify its entire public transport fleet show a very high level of ambition for a megacity.

In terms of future implementation, Istanbul is set to deploy advanced solutions such as Floating SPP systems on water reservoirs, massive battery storage for night-time energy stability, and neighborhood-based smart microgrids. Meanwhile, Alphen aan den Rijn is focusing its efforts on the "Programma energietransitie," which prioritizes building isolation to save energy and the development of sustainable business parks. Their most significant goal is the complete transition away from natural gas by the year 2050 to protect the climate.

The two cities can gain valuable insights from their differing approaches to the energy transition. Istanbul can learn the importance of energy isolation and insulation from Alphen, as reducing waste is often more effective than just generating more power. Conversely, Alphen aan den Rijn can learn from Istanbul's spatial efficiency; Istanbul's strategy of utilizing rooftops, parking lots, and water surfaces provides an excellent model for how Alphen can maximize solar energy production without using up valuable land.





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Alphen aan den Rijn

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